

Social Media Content Generation in Multiple Languages

BrandyAI — one line of input, Instagram-ready campaigns in seven Indian languages, on Microsoft AI Foundry.

Team

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Deliverables

1 — GITHUB REPOSITORY

<https://github.com/rudrakshxgupta/marketing-campaign-generator>

Python/FastAPI backend, React frontend, 269 passing tests. Setup instructions in README.md.

2 — WORKING PROTOTYPE

<https://github.com/rudrakshxgupta/marketing-campaign-generator#running-the-prototype>

Runs locally against Azure AI Foundry; it is not publicly hosted, because the deployment keys are personal and the image quota is small. Screenshots and results follow in the presentation below.

3 — 5-MINUTE DEMONSTRATION VIDEO

<https://youtu.be/FApW-GayaqE>

Narrated walkthrough, 1920×1080. Roughly half is unedited screen recording of the running application.

4 — PRESENTATION VIDEO

<https://youtu.be/64GgajEwDYE>

Slide-led walkthrough of the problem, the architecture and the results.

What it does

A marketer types one line — what they sell. A model compiles that into a creative brief, a single image call produces one **text-free** photograph, and the copy for every language is then typeset onto it and composited. The logo is placed from the brand's real file, never drawn by the model.

1 → 7

One image call produced seven finished deliverables in the recorded run. An eighth language costs nothing.

7 languages

English, Hindi, Hinglish, Marathi, Bengali, Tamil, Telugu — shaped correctly via HarfBuzz.

Transcreated

269 tests

Copy is authored per language, not translated: Diwali for Hindi, Durga Puja for Bengali, Pongal for Tamil.

Including subject-fidelity checks that flag a building whose floor count changed.

Built on Microsoft AI Foundry — FLUX.2-pro for imagery, GPT-5-mini for the brief and copy, headless Chromium for Indic typesetting.

PROJECT SUBMISSION

Social Media Content Generation in Multiple Languages

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PROBLEM STATEMENT

India advertises in English to an audience that does not read it.

Scale. Over 60 million small businesses market on Instagram. Fewer than 11% of Indians speak English; the market that converts reads Hindi, Bengali, Tamil, Telugu, Marathi.

Cost. An agency campaign in five languages runs to tens of thousands of rupees per creative set — out of reach for a single-shop seller.

The shortcut fails. Generating one English post and translating it afterwards produces copy that is accurate and dead, and carries the wrong cultural referent entirely.

A Diwali line translated into Bengali is still a Diwali line — when the campaign that audience responds to is Durga Puja.

Why existing tools do not solve it

Canva / templates

Human designs the layout and writes every language by hand. No generation, no scale, and Indic text is typed into a box that was never metric-tested for it.

ChatGPT + an image model

Produces an English image with lettering drawn *into* the picture. Ask for Devanagari and the model garbles it — plausibly enough that an English speaker ships it.

Ad platforms' auto-translate

Translates finished copy. The words change; the festival, the idiom and the offer framing do not.

None of them can put correct Tamil typography on a generated photograph at a price a small seller will pay. That is the gap.

OBJECTIVES

What the system had to achieve

1 — Minimal input

The user supplies only what they sell. The brief, the proposition, the occasion and the copy are all derived.

2 — Correct Indic typography

Conjuncts, matras and reordering right in every script — verified, not assumed.

3 — Culturally correct copy

Authored per language from a shared strategy, so the festival changes and not only the words.

4 — Economical

Adding a language must not cost another generation.

5 — Truthful

No invented claims, and a real subject must not be reinvented.

6 — Publishable

Instagram formats and safe zones enforced, not merely suggested.

Three measured platform facts, established first

Each was verified against the live platform before any code was written. Every structural decision that follows is forced by one of them.

Image models are English-only

They declare Languages: en. Diffusion garbles Indic scripts in a uniquely dangerous way — plausible to an English-speaking operator, illiterate to the audience.

Generation is rate-limited and billed

2–15 requests per minute. Seven languages across four formats as separate calls is neither affordable nor fast.

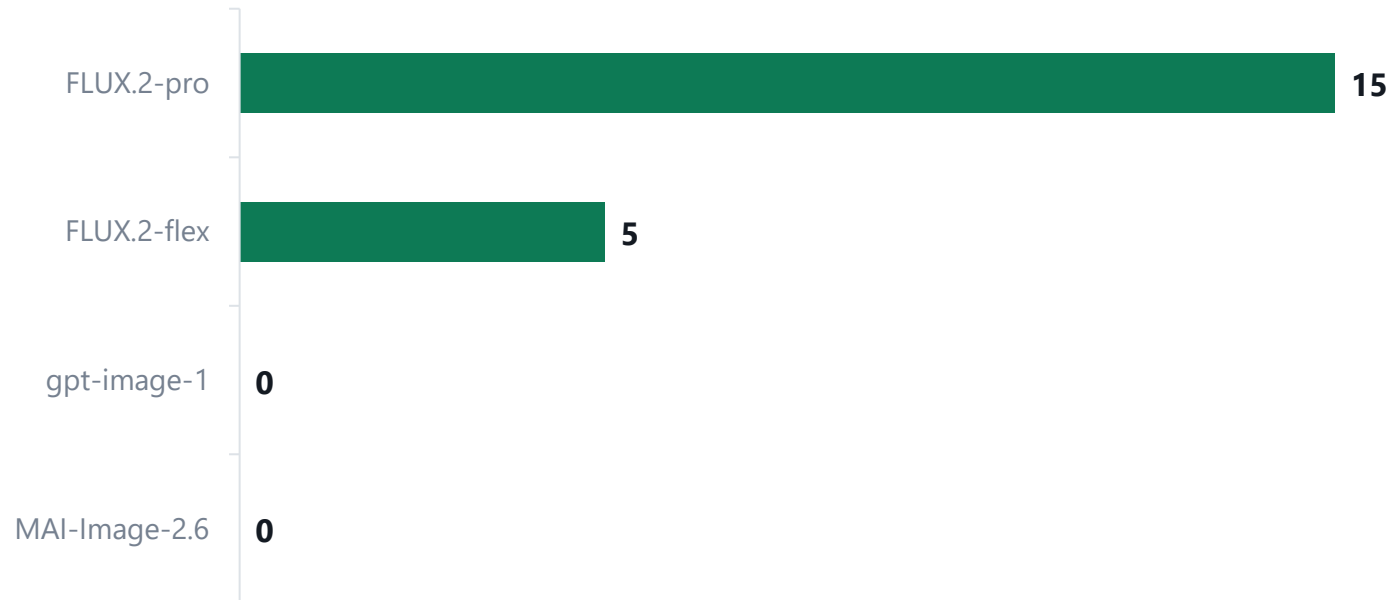
OCR cannot read the scripts

Azure Vision Read covers none of Bengali, Tamil, Telugu, Kannada, Malayalam, Gujarati, Gurmukhi or Odia. Model-drawn text could not be verified even if we wanted it.

All three point the same way — and that convergence is the design.

The model this was designed for had no quota

Reported by Azure itself, in every region tried. The project was built for MAI-Image-2.6; this is why it runs on FLUX.



“the quota limit is 0 for quota Requests Per Minute — MAI-Image-2.6”

Both image backends already implemented one interface, so switching cost a single configuration line and nothing downstream learned which model was in use.

PROPOSED SOLUTION

Generate one text-free image. Composite every language on top.

The image model never draws text. It is asked for a photograph with a named region deliberately left clean.

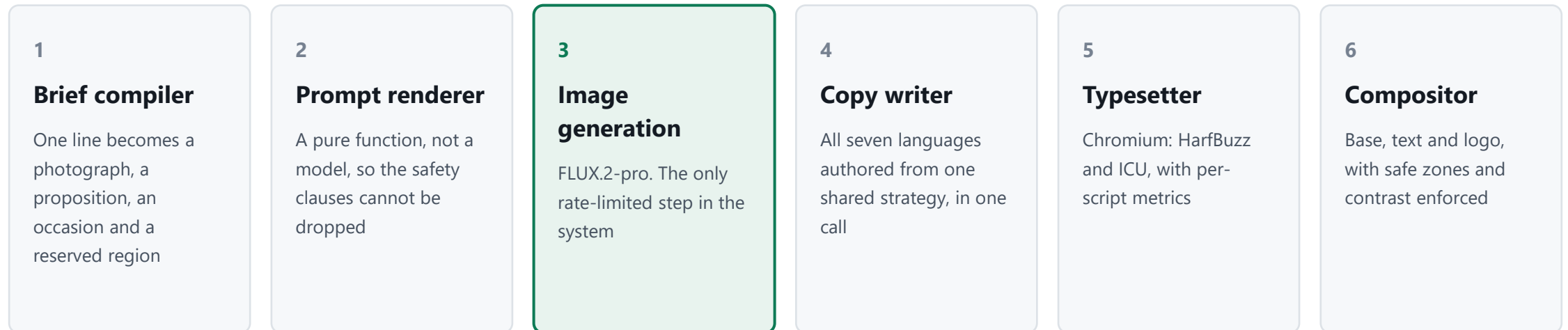
Typography is typeset, not generated. Headless Chromium carries HarfBuzz and ICU — the same shaping engine a browser uses, continuously tested against these scripts by a browser vendor.

The logo is composited from the real file. A generated logo is a wrong logo.

Consequences

The seventh language costs zero model calls · the logo is byte-exact rather than hallucinated · QA for the text path becomes arithmetic instead of inference.

Pipeline

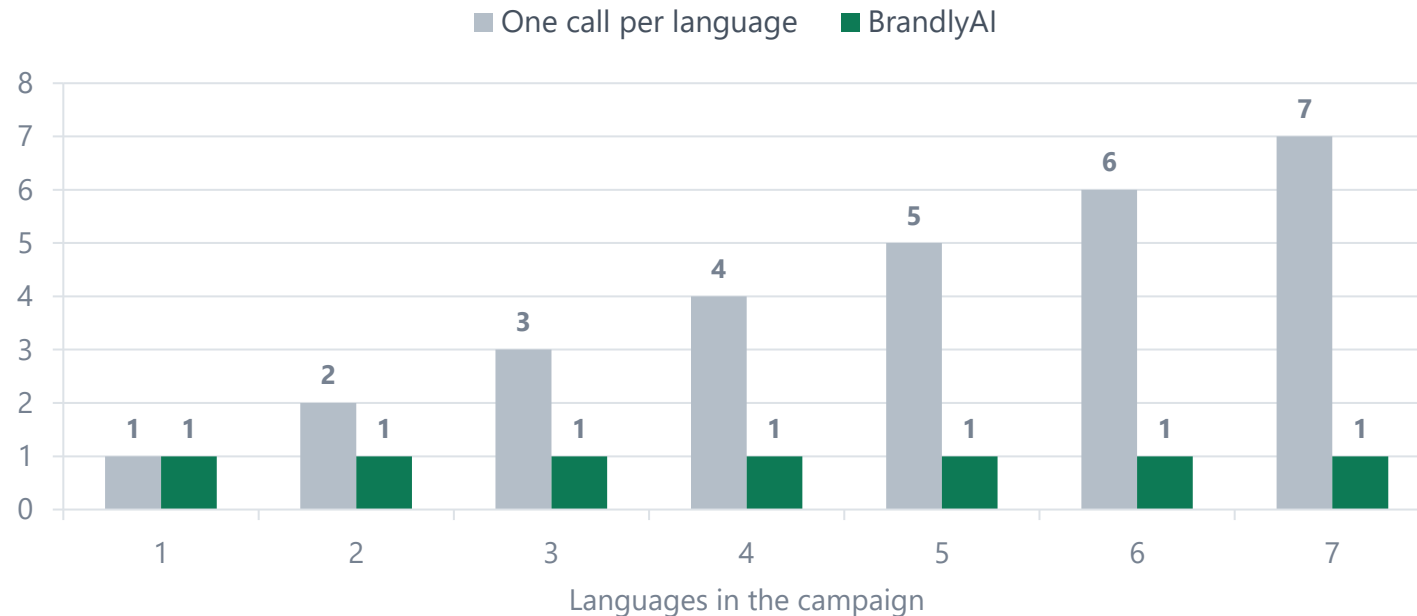


Only stage 3 touches the image quota. Everything after it is CPU work that scales freely — which is precisely what makes the seventh language free and the seventh format cost a generation.

COST MODEL

Adding a language costs nothing

Image calls required to produce a campaign in N languages. The gap between the two series is the entire architectural argument.



Languages

are typeset onto the one generated photograph, so the seventh costs nothing and the fiftieth would too.

Formats

do cost a generation each — they are different pictures. Economy mode crops them from one tall master when that trade is worth making.

Technology stack

Microsoft AI Foundry

FLUX.2-pro for imagery, GPT-5-mini for the brief and copy. Both image backends implement one interface, so the model is a configuration line.

Python · FastAPI

Async job API. Generation is never synchronous: a request that waited for an image would time out behind almost any proxy.

Headless Chromium · Playwright

Indic typesetting. Pillow on this machine has no RAQM and would emit silently broken glyph order.

Pillow

Cropping, upscaling and compositing only — never text.

React · Vite

Instagram phone-frame preview with the platform's real safe zones drawn to scale.

pytest

269 tests, including fidelity measurement and shaping proofs.

KEY CONTRIBUTIONS

What makes it more than a wrapper

Transcreation, not translation

Copy is authored per language from a language-neutral strategy, so the referent changes: Diwali for Hindi, Durga Puja for Bengali, Pongal for Tamil — from one campaign.

Measured subject fidelity

An uploaded building is restaged, never redesigned. Outline overlap and detail density are measured against the original; architecture is held to a stricter floor than food.

Contrast-aware compositing

The logo's light or dark version is chosen by measuring the pixels behind it, scored against the worst tone in the patch rather than the average.

Image-driven layout

Eight candidate regions are scored for edge energy and tonal spread, so copy lands where the picture is actually calm.

Enforced spend ceiling

Checked before the call, persisted to disk, with a cache that serves repeats without touching the budget.

Free human review

The base carries no text, so corrections are a typesetting pass. The creative updates as you type.

RESULTS

One image call, seven finished deliverables



English



Hindi



Bengali



Tamil

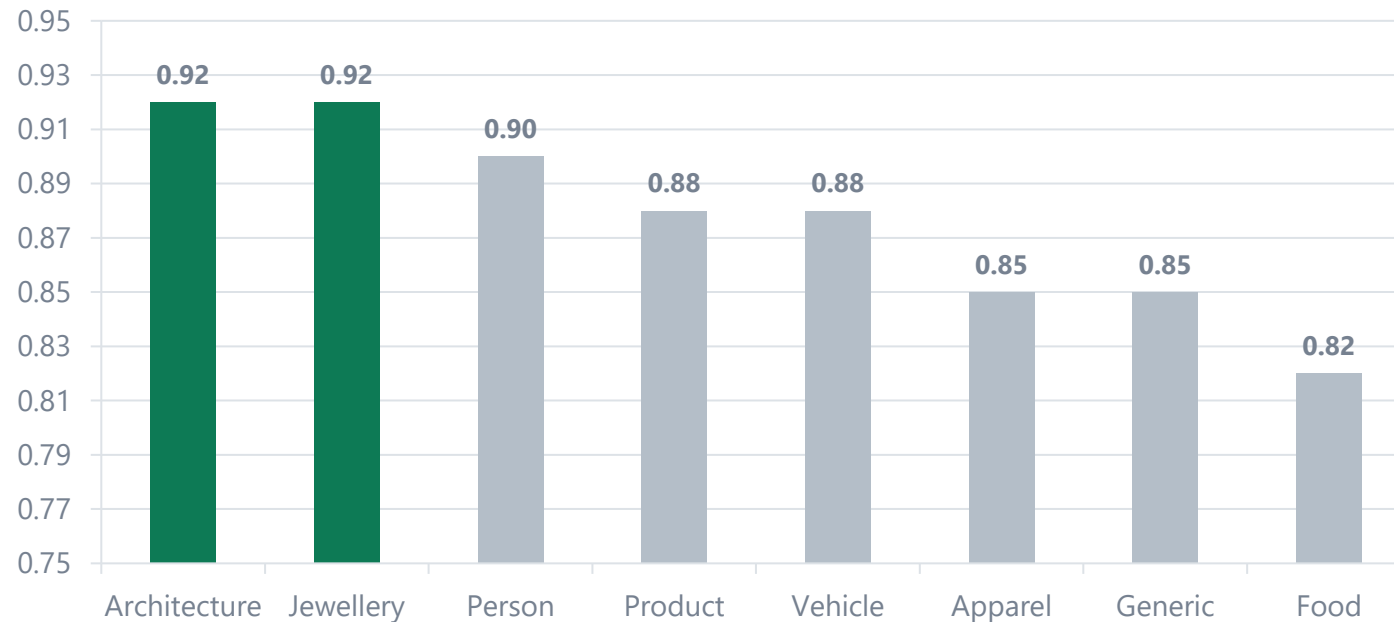


Telugu

The same photograph underneath every one. The text is typeset and composited — never drawn by the model.

How much drift each subject is allowed

Similarity floors from app/copy/fidelity.py. A result below its floor is flagged for review rather than published.



Why architecture is strictest

A building that gains a storey or loses a window renders beautifully and advertises a property that does not exist. That is a legal exposure, and it is invisible to whoever is running the tool.

A slightly different plate of food is not. The scale reflects consequence, not difficulty.

VALIDATION

Measured, not asserted

1 → 7

image calls to finished deliverables,
in the recorded run

269

automated tests, including shaping
proofs and fidelity checks

7

locales: English, Hindi, Hinglish,
Marathi, Bengali, Tamil, Telugu

5

Instagram formats with safe zones
enforced as a hard gate

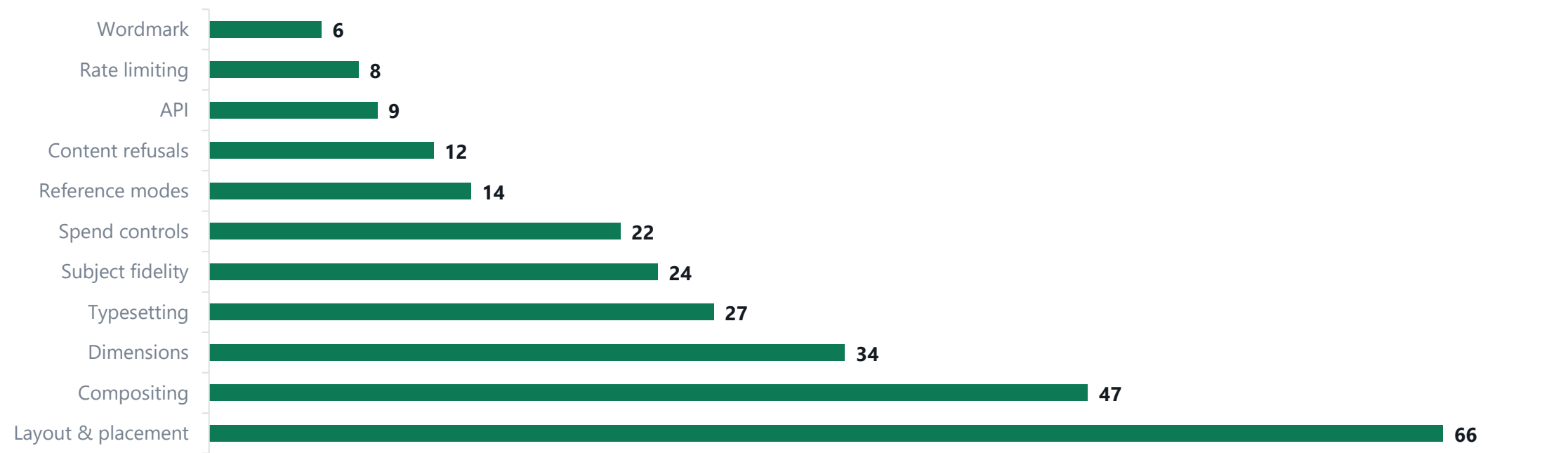
Shaping is proven, not hoped for. A test compares the rendered width of a conjunct against the same characters forced apart with a zero-width non-joiner. Without shaping the two widths match.

Fidelity is measured against the original. Synthetic buildings that gain a storey or a window are detected; a recoloured sky is not flagged, because only the subject is compared.

TEST COVERAGE

Where the 269 tests are

Collected test count by area. Weighted towards the parts that fail silently — layout, compositing and typography all produce output that looks plausible when wrong.



CHALLENGES

Problems met, and how they were solved

Every Microsoft image model had zero quota

Azure's own error: "the quota limit is 0". Both backends already implemented one interface, so switching to FLUX.2-pro cost a single configuration line.

Our own safety clause was being refused

A blocklist matches terms and cannot read negation, so "no national flag" reads as a prompt about flags. Constraining the frame positively excludes the same categories without naming them.

A single word broke every menswear brief

"men's" is on the provider blocklist. Found by probing the live filter — identical sentences, one word apart — and now stripped with a four-rung recovery ladder behind it.

Each was diagnosed by measurement rather than guesswork; the probe that found the blocked word cost three images and settled a question two days of reading had not.

What is not built — stated plainly

Claiming completeness that does not exist would be the least defensible thing in this report.

Content safety. Foundry provides no built-in filtering for FLUX, so output screening is entirely ours and is not yet implemented.

Provenance and AI disclosure. MeitY's IT Amendment Rules took effect 20 February 2026; synthetic visual content must be labelled. C2PA signing is designed but not built.

OCR zero-text gate. The base is asked to contain no text; that is not yet verified automatically for the scripts OCR can read.

Category compliance. RERA numbers for real estate, FSSAI for food, and unsubstantiated superlatives in every language.

Persistence. Jobs live in memory and do not survive a restart; fonts are not yet bundled and pinned.

CONCLUSION

One architectural decision carries the project

Generating one text-free image and compositing every language on top is not an optimisation. It is what makes correct Indic typography possible at all, and it is what makes the seventh language free.

Three measured platform constraints pointed at the same design. The system that follows is cheaper, verifiable, and honest about what it does not yet do.

Repository

github.com/rudrakshgupta/marketing-campaign-generator